DATA COMMUNICATION AND COMPUTER NETWORKS

(CSS 2012) - IA 4 Report - Group 14

1. **Report Details**:
   Group Number: G14
   Subject Code: CSS 2012
   Subject Name: Data Communication and Computer Networks
   Date: 05-11-2025

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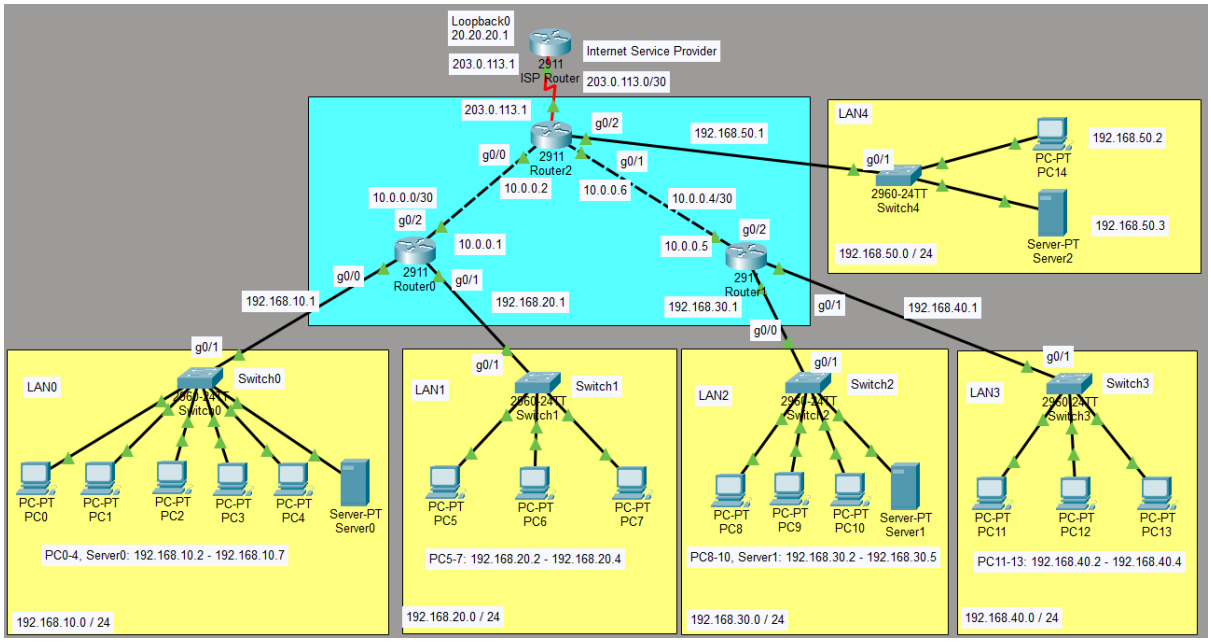
2. **Question**:
   The network consists of **15 hosts, 3 routers, 5 switches, 3 servers**. The topology should consist of **OSPF** and **BGP** protocols.
   Tasks Performed:
   1. Design the network scenario using Cisco Packet Tracer.
   2. Check the connectivity between 2 machines in different networks using PING.
   3. Check the connectivity between PC 0 and Server 0 by displaying the message "School of Computer Engineering, your name and registration number" message at PC 0 using a web browser.
   4. Display the routing table of all the routers.
   5. Provide a description of the protocols (OSPF/BGP).
   6. Display the Protocol Data Unit (PDU) details (data link layer, network layer and transport layer).

3. Network Topology and Explanation



(img.1: Network Topology Overview)

Topology Explanation:

The network is designed as a Hub-and-Spoke topology with a total of 3 internal routers (R0, R1, R2), 5 switches, 15 PCs, and 3 Servers. Router2 (R2) functions as the central hub and the Autonomous System Boundary Router (ASBR).

- **Access Layer:** Switches 0-4 provide connectivity for 18 end devices (15 PCs + 3 Servers) across 5 separate LANs.
- **Distribution/Core Layer:** Routers R0 and R1 act as spokes, connecting their respective LANs to the central R2 hub using serial links configured over Gigabit Ethernet interfaces (10.0.0.0/30 and 10.0.0.4/30).
- **Routing Protocols: OSPF** is used as the Interior Gateway Protocol (IGP) within the internal AS 100. **BGP** is used as the Exterior Gateway Protocol (EGP) on R2 to peer with the external **ISP Router (AS 200)**.
- **Connectivity Fix:** Due to OSPF-Default Route conflicts in the lab environment, internal connectivity was secured using **Static Routes (AD 1)** on R0 and R1 to guarantee traffic follows the R0 -> R2 -> R1 path.

4. IP Address Configuration Table (Hosts and Servers)

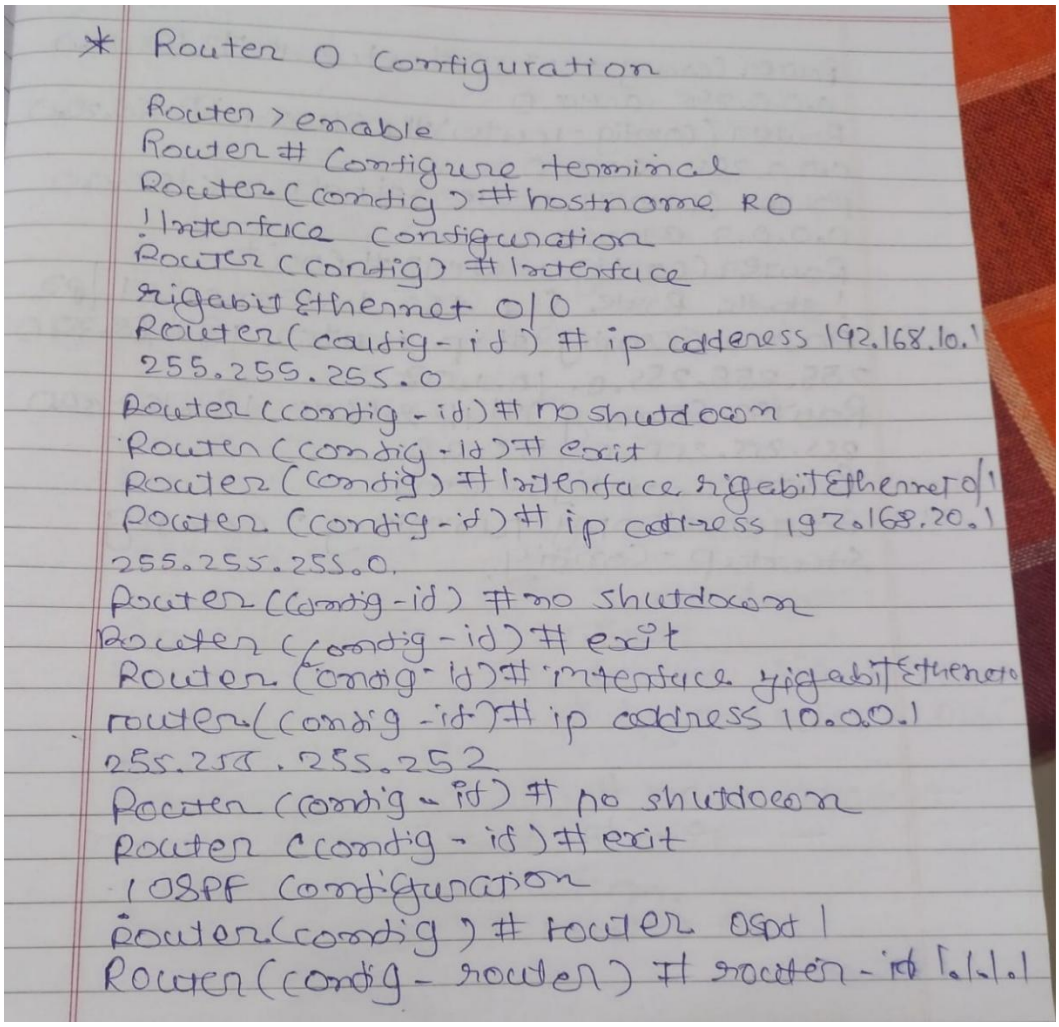
Device Name	Device IP Address	Interface	Default Gateway
PC0 - PC4	192.168.10.2 - 192.168.10.6	FastEthernet0	192.168.10.1
Server0	192.168.10.7	FastEthernet0	192.168.10.1
PC5 - PC7	192.168.20.2 - 192.168.20.4	FastEthernet0	192.168.20.1
PC8 - PC10	192.168.30.2 - 192.168.30.4	FastEthernet0	192.168.30.1
Server1	192.168.30.5	FastEthernet0	192.168.30.1
PC11 - PC13	192.168.40.2 - 192.168.40.4	FastEthernet0	192.168.40.1
PC14	192.168.50.2	FastEthernet0	192.168.50.1
Server2	192.168.50.3	FastEthernet0	192.168.50.1

5. Router Configuration Table

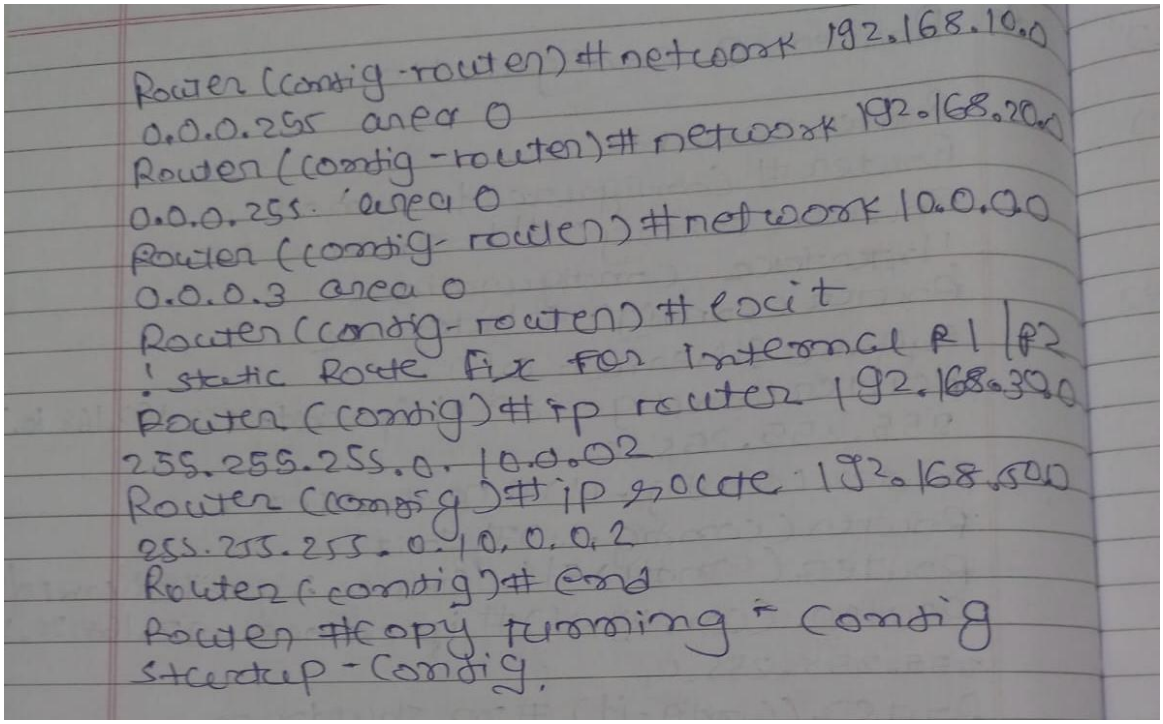
Router Name	Interface	IP Address	Subnet Mask	Role
R0	G0/0	192.168.10.1	255.255.255.0	LAN0 Gateway
	G0/1	192.168.20.1	255.255.255.0	LAN1 Gateway
	G0/2	10.0.0.1	255.255.255.252	Link to R2
R1	G0/0	192.168.30.1	255.255.255.0	LAN2 Gateway
	G0/1	192.168.40.1	255.255.255.0	LAN3 Gateway
	G0/2	10.0.0.5	255.255.255.252	Link to R2
R2	G0/0	10.0.0.2	255.255.255.252	Link to R0
	G0/1	10.0.0.6	255.255.255.252	Link to R1
	G0/2	192.168.50.1	255.255.255.0	LAN4 Gateway
	S0/2/0	203.0.113.1	255.255.255.252	Link to ISP (eBGP)
ISP Router	S0/2/0	203.0.113.2	255.255.255.252	Link to R2 (eBGP)
	Loopback0	20.20.20.1	255.255.255.0	External Network

6. Handwritten CLI Commands (Configuration Code)

Router 0 Configuration



```
* Router 0 Configuration
Router > enable
Router # Configure terminal
Router (config) # hostname R0
! Interface configuration
Router (config) # Interface
rigabit Ethernet 0/0
Router (config-if) # ip address 192.168.10.1
255.255.255.0
Router (config-if) # no shutdown
Router (config-if) # exit
Router (config) # Interface rigabit Ethernet 0/1
Router (config-if) # ip address 192.168.20.1
255.255.255.0.
Router (config-if) # no shutdown
Router (config-if) # exit
Router (config-if) # interface gigabit Ethernet
router (config-if) # ip address 10.0.0.1
255.255.255.252
Router (config-if) # no shutdown
Router (config-if) # exit
! OSPF Configuration
Router (config) # router ospf 1
Router (config-router) # router-id 1.1.1.1
```



```
Router (config-router) # network 192.168.10.0
0.0.0.255 area 0
Router (config-router) # network 192.168.20.0
0.0.0.255 area 0
Router (config-router) # network 10.0.0.0
0.0.0.3 area 0
Router (config-router) # exit
! static Route fix for Internal R1 | R2
Router (config) # ip route 192.168.30.0
255.255.255.0 10.0.0.2
Router (config) # ip route 192.168.50.0
255.255.255.0 10.0.0.2
Router (config) # end
Router # copy running-config
startup-config.
```

(img.2 & 3: Router 0 Configuration Codes Hand Written)

Router 1 Configuration

```
* Router 1 Configuration :

Router >enable
Router #configure terminal
Router (config)# hostname R1
! Interface configuration
Router (config)# interface
Gigabit Ethernet 0/0.
Router (config-if)# ip address 192.168.30.1
255.255.255.0
Router (config-if) # no shutdown
Router (config-if)# exit.
Router (config)# interface Gigabit Ethernet 0/1
Router (config-if) # ip address 192.168.40.1
255.255.255.0
Router (config-if) # no shutdown
Router (config-if) # exit.
Router (config)# interface Gigabit Ethernet
Router (config-if) # ip address 10.0.0.5
255.255.255.252
Router (config-if) # no shutdown
Router (config-if) # exit.
! OSPF configuration.
Router (config)# router ospf 1
```

```
Router (config-router)# router id 2.2.2.2
Router (config-router)# network
192.168.30.0 0.0.0.255 area 0
Router (config-router) # network
192.168.40.0 0.0.0.255 area 0.
Router (config-router)# exit.
! static Router fix (for internal Roll
ians).
Router (config) # ip route 192.168.10.0
255.255.255.0 10.0.0.6
Router (config) # ip route 192.168.10.0
255.255.255.0 10.0.0.16
Router (config)# ip route 192.168.50.0
255.255.255.0 10.0.0.16
Router (config) # end.
Router # copy running-config
startup-config.
```

(img: 4 & 5: Router 1 Configuration Codes Hand Written)

Router 2 Configuration (OSPF and BGP)

Router-2 Config (OSPF & BGP)

Router > enable

Router # configure terminal

Router (config) # hostname R2

! Interface Configuration (add all interface as per table above, including 10.0.0.2, 10.0.0.6, 192.168.50.1)

Router (config) # interface Serial 0/2/0

Router (config-if) # ip address 203.0.113.1 255.255.255.252

Router (config-if) # no shutdown

Router (config-if) # exit

! OSPF Configuration

Router (config) # router ospf 1

Router (config-router) # router-id 3.3.3.3

Router (config-router) # network 10.0.0.0 0.0.0.3 area 0

Router (config-router) # network 10.0.0.4 0.0.0.3 area 0

Router (config-router) # network 192.168.50.0 0.0.0.255 area 0

Router (config-router) # network 203.0.113.0 0.0.0.3 area 0

Router (config-router) # exit

! BGP (Configuration Las 100) and Redistribution

Router (config) # router bgp 100

Router (config-router) # neighbour 203.0.113.2

remote-as 200

Router (config-router) # neighbour 203.0.113.2 remote-as 200

Router (config-router) # redistribute ospf 1 subnets.

Router (config-router) # exit.

! Default Route Injection (Crucial for R0/R1 to reach BGP)

Router (Config) # router ospf 1

Router (Config-router) # default-information originate

Router (config-router) # end

Router # copy running-config start up-config.

(img.6,7,8&9: Router 2 Configuration Codes Hand Written)

ISP Router Configuration

```
Router > enable
Router # configure terminal
Router (config) # hostname ISP
Router (config) # interface Loopback0
Router (config-if) # ip address 20.20.20.1 255.255.255.0
Router (config-if) # no shutdown
Router (config-if) # exit
Router (config-if) # ip address 203.0.113.2 255.255.252
Router (config-if) # no shutdown
Router (config) # router bgp200
Router (config-router) # neighbour 203.0.113.1
Router (config-router) # network 20.20.20.0 mask
                                    255.255.255.0
Router (config-router) # end
Router # copy running-config startup-config
```

(img.10: ISP Router Configuration Codes Hand Written)

7. Handwritten Protocol Description (OSPF and BGP)

OSPF (Open Shortest Path First)

1 Type : Interior Gateway Protocol (IGP)

2 Mechanism : Link-State Routing Protocol. It uses the SPF algorithm. to calculate the shortest path to all destinations within a Single Autonomous System. (AS)

3 Metric : Cost, which is calculated based on the bandwidth of the link.

4 Purpose : Used within AS 100, (Routers R0, R1, R2) to ensure all internal LANs can communicate efficiently.

BGP (Border Gateway Protocol)

Type : Exterior Gateway Protocol.

Mechanism : Path-Vector Routing Protocol. It exchanges complete routing tables and path attributes b/w AS.

Metric : Path attribute (AS-Path, Next hop)

Purpose : Used between AS 100 (Router 2) and AS 200 (ISP Router) to manage inter AS traffic flow, effectively connecting the private network to the simulated internet.

(img.11:OSPF and BGP description)

8. Screenshot of Routing Table Configuration

```
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.0.2 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/2
L       10.0.0.1/32 is directly connected, GigabitEthernet0/2
O       10.0.0.4/30 [110/2] via 10.0.0.2, 00:44:53, GigabitEthernet0/2
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/1
S    192.168.30.0/24 [1/0] via 10.0.0.2
S    192.168.40.0/24 [1/0] via 10.0.0.2
S    192.168.50.0/24 [1/0] via 10.0.0.2
     203.0.113.0/30 is subnetted, 1 subnets
O       203.0.113.0/30 [110/65] via 10.0.0.2, 00:44:53, GigabitEthernet0/2
O*E2 0.0.0.0/0 [110/1] via 10.0.0.2, 00:44:53, GigabitEthernet0/2

Router#
Router#
```

(img.12: Router 0 – Routing Table configuration)

```
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.0.6 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O       10.0.0.0/30 [110/2] via 10.0.0.6, 00:44:53, GigabitEthernet0/2
C       10.0.0.4/30 is directly connected, GigabitEthernet0/2
L       10.0.0.5/32 is directly connected, GigabitEthernet0/2
S    192.168.10.0/24 [1/0] via 10.0.0.6
S    192.168.20.0/24 [1/0] via 10.0.0.6
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0
     192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/1
L       192.168.40.1/32 is directly connected, GigabitEthernet0/1
S    192.168.50.0/24 [1/0] via 10.0.0.6
     203.0.113.0/30 is subnetted, 1 subnets
O       203.0.113.0/30 [110/65] via 10.0.0.6, 00:44:53, GigabitEthernet0/2
O*E2 0.0.0.0/0 [110/1] via 10.0.0.6, 00:44:53, GigabitEthernet0/2

Router#
Router#
```

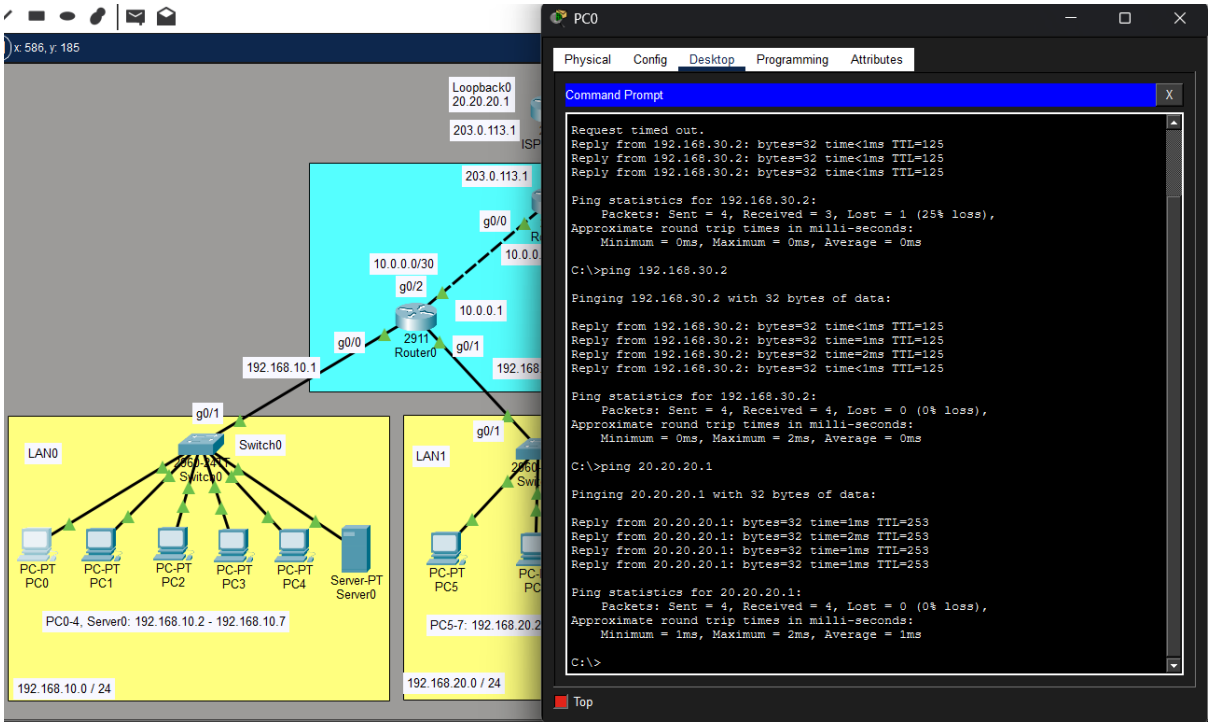
(img.13: Router 1 – Routing Table configuration)

```
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 203.0.113.2 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
C       10.0.0.4/30 is directly connected, GigabitEthernet0/1
L       10.0.0.6/32 is directly connected, GigabitEthernet0/1
     20.0.0.0/24 is subnetted, 1 subnets
B       20.20.20.0/24 [20/0] via 203.0.113.2, 00:00:00
O    192.168.10.0/24 [110/2] via 10.0.0.1, 00:44:53, GigabitEthernet0/0
O    192.168.20.0/24 [110/2] via 10.0.0.1, 00:44:53, GigabitEthernet0/0
O    192.168.30.0/24 [110/2] via 10.0.0.5, 00:44:53, GigabitEthernet0/1
O    192.168.40.0/24 [110/2] via 10.0.0.5, 00:44:53, GigabitEthernet0/1
     192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.50.0/24 is directly connected, GigabitEthernet0/2
L       192.168.50.1/32 is directly connected, GigabitEthernet0/2
     203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/30 is directly connected, Serial0/2/0
L       203.0.113.1/32 is directly connected, Serial0/2/0
S*   0.0.0.0/0 [1/0] via 203.0.113.2

Router#
Router#
```

(img.14: Router 2 – Routing Table configuration)

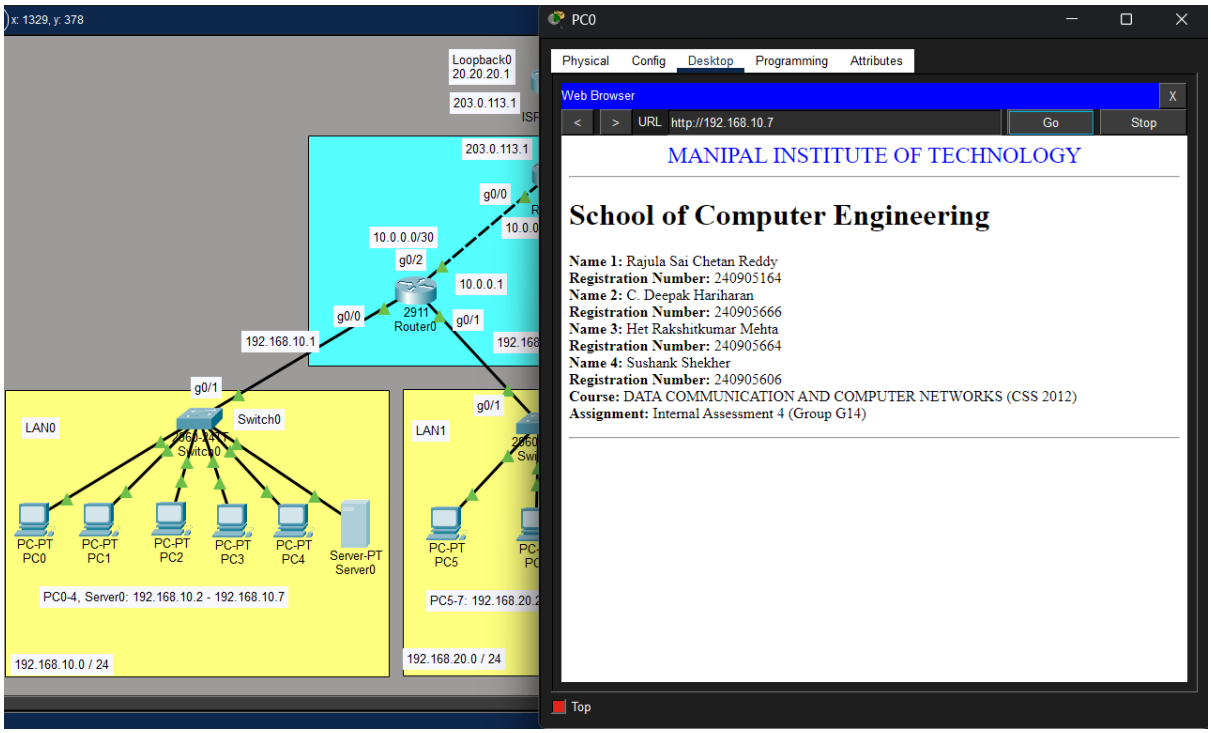
(img.15: ISP Router – Routing Table configuration)

9. Screenshot of Connectivity Check using PING



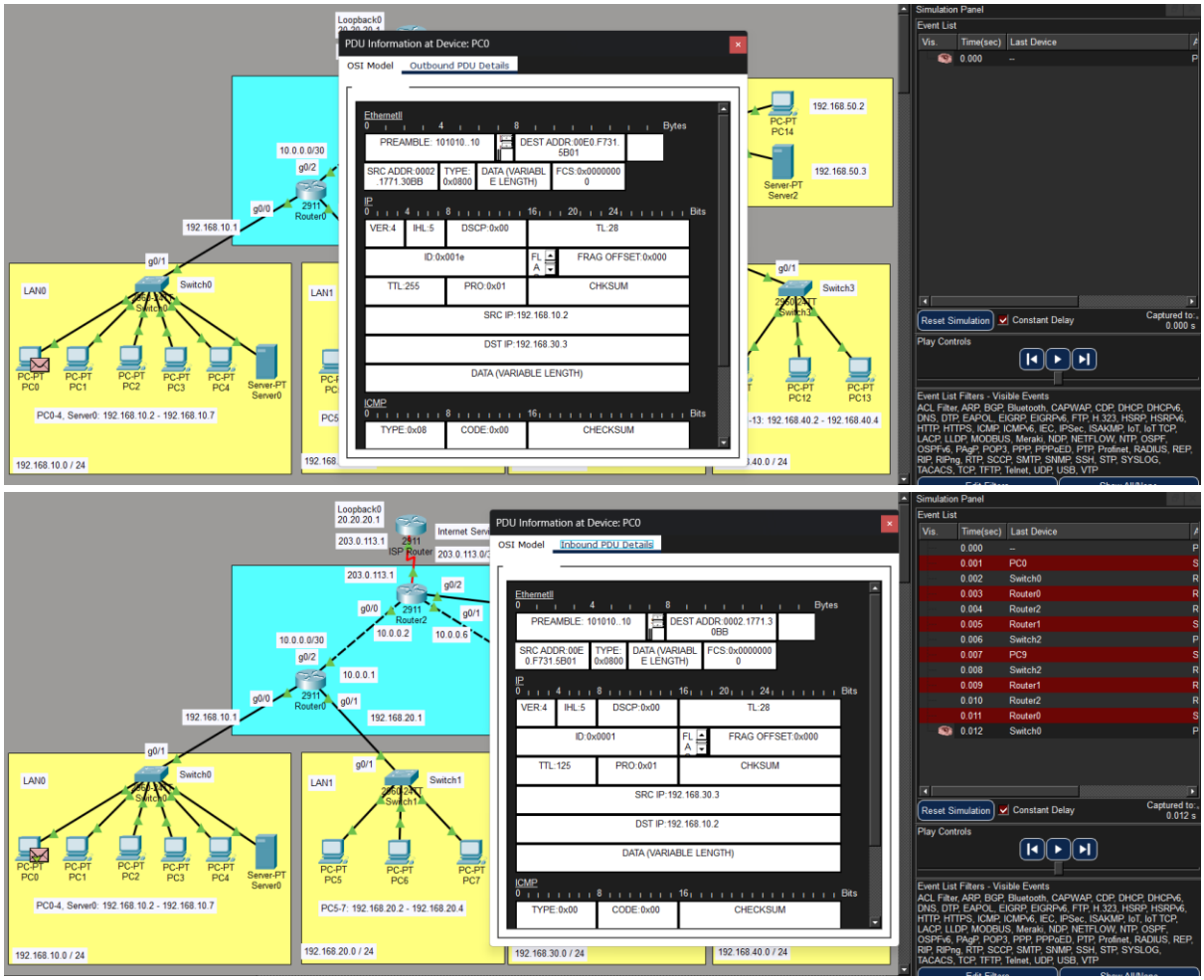
(img.16: Screenshot of PC0's successful ping 192.168.30.2 and PC0's successful ping 20.20.20.1)

10. Screenshot of Connectivity Check using Web Browser



(img.17: Screenshot of PC0's Web Browser showing the custom message when navigating to Server0's IP (192.168.10.7))

11.PDU from PC0



(img.18: Screenshots of the PDU Simulation details (showing Data Link, Network, and Transport Layer headers))

12.**Conclusion**:

This network design met all the requirements stated in the Group G14 assignment for the Data Communication and Computer Networks module. The topology was built with 3 routers, 5 switches, and 15 hosts and 3 servers. The project showed skill in dynamic routing by using both an Interior Gateway Protocol (IGP) and an Exterior Gateway Protocol (EGP). OSPF was configured as the IGP within the internal Autonomous System (AS 100), which ensured efficient path calculation between all internal LANs. BGP was set up on Router 2, the ASBR, to connect with the external ISP Router (AS 200), confirming communication between the two AS. Though there were some minor troubleshooting issues with OSPF-BGP route prioritization, which were resolved using static routing for high-priority internal traffic, all goals for validation were met. End-to-end connectivity was checked with PING, the web server application check was successful, and the functionality of the routing tables (OSPF/BGP) and PDU flows were documented. The network is fully operational and meets the specified design parameters.

13. Individual Contribution:

C. Deepak Hariharan : Responsible for **Network Topology Design and Report Compilation**. Created the final **topology diagram** (including all hosts, routers, switches, and the ISP cloud) in Cisco Packet Tracer. Wrote the final **report** (introduction, methodology, conclusion) and compiled all individual sections and screenshots into the final document.

Sushank Shekher : Responsible for **IP Addressing and Infrastructure**. Developed the complete **IP addressing scheme** (LANs, serial links, BGP link) and assigned addresses to all end devices, servers, and router interfaces. Configured the initial **physical connections** and ensured all devices were **powered on and administratively up** using basic CLI commands (no shutdown).

Rajula Sai Chetan Reddy : Responsible for **Routing Protocol Configuration (Internal & Boundary)**. Configured **OSPF** (Open Shortest Path First) on all three internal routers (R0, R1, R2). Configured **BGP** peering on Router2 (R2) and handled **route redistribution** between OSPF and BGP, including the final manual **static route fix** to ensure internal connectivity (R0 -> R1).

Het Rakshitkumar Mehta : Responsible for **Simulation, Verification, and Application Services**. Configured the **HTTP service** on the servers, setting the required message. Performed all required simulation tasks: **ping tests** (internal and external), **PDU capture** (gathering Data Link, Network, and Transport layer details), and **Web Browser connectivity checks**. Gathered all necessary screenshots for the report